

Q-1 Select the correct option from the following questions.

8

- (1) _____ is the developer of C++.
- A Dannis Ritchie B James Gosling
C Ivan Byros D Bjarne Strastrup
- (2) The header file _____ should be included that use i/o statements.
- A iostream B stdio
C conio D io
- (3) _____ is a linear data structure.
- A Stack B Tree
C Graph D None of these
- (4) _____ is/are the types of constructor.
- A Default B Copy
C Parameterized D All of Above
- (5) In row major order, elements of matrix are stored on a _____ basis.
- A Column by Column B Row by Row
C Row by Column D Column by Row
- (6) An array is a _____ data structure.
- A Heterogeneous B Homogeneous
C Non- Linear D Unordered
- (7) The terms "PEEP" and "CHANGE" are related to _____ Data Structure.
- A Queue B Stack
C Array D Linked List
- (8) Node which does not have any child node is called _____.
- A Root node B Leaf Node
C Sibling Node D Parent Node

[P.T.O.]

Q-2 Do as directed. (ATTEMPT ANY SEVEN) 14

- (1) Define: Data Members and Member functions.
- (2) List names of operators that cannot be overloaded in C++.
- (3) What is the difference between private & protected members of the class?
- (4) What is an operator function? Describe the syntax of an operator function.
- (5) Explain destructors with an example.
- (6) Differentiate: Binary Tree, Height of a Tree.
- (7) Define: File, Record, Field, Tree
- (8) What is File organization? What are their types?
- (9) Differentiate between primitive and non-primitive data types.

Q-3 A Write a note on new and delete operator. 6
B What is Constructor? List out them. Explain ANY ONE in detail. 6

OR

B What is Inheritance? List out them. Explain ANY ONE in detail. 6

Q-4 A Write a note on Unary operator overloading. 6
B Discuss function overloading with an example. 6

OR

B Explain Exception Handling in C++. 6

Q-5 A Write an algorithm on PUSH and POP operations of a stack. 6
B Write an algorithm to insert and delete an element in Queue. 6

OR

B Explain: Row and Column major order method for 2-D array with an example. 6

Q-6 A What is Hashing functions? List them. Explain any two with an example. 6
B Write a note on Sequential file. 6

OR

B Explain ISAM in detail. 6
